# Dynamics and Reconstruction from Observation

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**Abstract**

Starting from a $\sigma$-additive probability measure on an observable algebra [5], this paper derives two structures that are often assumed: the dynamics of prediction, and the recoverability of the state space.

Making prediction precise forces the introduction of the predictive kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$, the regular conditional distribution of a future observation $F$ given a current observation $Q$. Composing with $Q$ gives the minimal predictive state map $Q_* : \Omega \to \mathcal{P}(\beta)$. Indexing over time, the kernels compose via Chapman–Kolmogorov — derived, not assumed — yielding a Markov semigroup $\{K_t\}$ and the Koopman–Perron duality $\int K_t g \, d\mu = \int g \, d(\mu \star \Pi_t)$. When the kernels are Dirac measures, $K_t$ collapses to the classical Koopman operator.

The construction produces a predictive state space $\mathcal{P}(\beta)$. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$, the map $Q_*$ becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. The reconstruction question is then internal to the same framework: whether this predictive state is faithful — whether the prediction machinery sees all of $X$, or only a proper quotient.

The answer is a three-way equivalence — the reconstruction theorem. State-space recovery ($\mathcal{O}_h = \mathcal{B} \bmod \mu$) holds if and only if the delay algebra $\mathcal{A}_h$ is $L^2$-dense, if and only if $\Phi_h$ is a measure-theoretic embedding. The bridge between the first two conditions is a general lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty$ containing 1, the $L^2$-closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. When reconstruction holds, the compact Stone space [5] — built as a completion of the sample space for deriving $\sigma$-additivity — is identified with $X$ itself.

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## 1 Introduction

The classical approach to dynamics, following Koopman [3], describes how observable functions evolve under a known transformation $T$: the operator $U_T f = f \circ T$ acts by precomposing with $T$ as a primitive. This paper asks two questions that precede this picture.

The first is: given a current observation $Q : \Omega \to \alpha$ and a future observation $F : \Omega \to \beta$, what does an observer predict about $F$ given $Q$? The dynamics, if there are any, should follow from the answer — not precede it. Making prediction precise immediately requires a $\sigma$-additive probability measure on the observable algebra. That measure is not an extra assumption: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion [5], the condition ruling out persistent mass on globally vanishing events.

Once the measure exists, the canonical answer to the prediction question is the regular conditional distribution of $F$ given $Q$: the unique Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem [6], which requires $\sigma$-additivity. Composing with $Q$ gives the minimal predictive state map $Q_* : \Omega \to \mathcal{P}(\beta)$, the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators $\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The construction produces a predictive state space $\mathcal{P}(\beta)$ and a stationary distribution $\nu_{Q_*}$. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$ for some $h \in L^\infty$, the map $Q_*$ specialises to the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$. The reconstruction question is therefore not a separate problem: it asks whether this predictive state is faithful — whether the prediction machinery sees all of $X$, or only a proper quotient. The answer is a three-way equivalence: the $\sigma$-algebraic condition $\mathcal{O}_h = \mathcal{B}$ modulo $\mu$, the analytic condition that the delay algebra $\mathcal{A}_h$ is $L^2$-dense, and the geometric condition that $\Phi_h$ is a measure-theoretic embedding are all equivalent.

The key to the equivalence is the density bridge lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$ containing the constant 1, the $L^2$-closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. The proof goes via the functional monotone class theorem [1]. This is what lets the $\sigma$-algebraic condition ($\mathcal{O}_h = \mathcal{B}$ mod $\mu$) and the analytic condition ($\mathcal{A}_h$ $L^2$-dense) pass freely between each other — they are not two separate conditions but two descriptions of the same constraint.

When reconstruction holds, the Stone space [5] $\mathrm{St}(\mathcal{O}_h)$ — built as a compact completion of the sample space for deriving $\sigma$-additivity — turns out to be measure-theoretically isomorphic to $X$ itself. The programme that began by seeking a compact ambient space for the measure ends by identifying it with the state space being observed.

The paper is self-contained modulo the probability measure. Sections 2 and 3 develop the predictive structure from a probability space $(\Omega, \mathcal{F}, P)$. Section 4 makes the connection explicit: specialising $Q_*$ to a measure-preserving system yields the delay map, and the reconstruction question becomes whether that map is injective modulo null sets. Sections 5 and 6 carry out the reconstruction, proving the density bridge and the reconstruction theorem.

# 2 Prediction and the minimal predictive state

## 2.1 Predictive kernel

Fix a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on $\beta$ is used once: it is the hypothesis under which the Rokhlin disintegration theorem [6] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on $\beta$, equipped with its canonical (Giry) $\sigma$-algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of $F$ given $Q$ is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\ F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

**Definition 2.1** (Predictive kernel)**.** The *predictive kernel* is $\Pi_Q$, the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \to \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$-almost every $q$, by uniqueness of disintegration.

Composing $\Pi_Q$ with $Q$ gives the map sending each sample point to its conditional law.

### 2.2 Minimal predictive state

**Definition 2.2** (Minimal predictive state map)**.** The *minimal predictive state map* is

$$Q_* : \Omega \to \mathcal{P}(\beta), \qquad Q_*(\omega) \;=\; \Pi_Q(Q(\omega), \cdot).$$

Since $\Pi_Q$ is a Markov kernel it is measurable, and $Q_*$ inherits measurability by composition.

**Theorem 2.3** (Predictive factorization)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \int g \, dQ_*(\cdot) \quad P\text{-a.e.}$$

*Proof.* Let $h(\omega) := \int g \, dQ_*(\omega)$. We verify the two defining properties.

*Measurability.* The map $q \mapsto \int g \, d\Pi_Q(q, \cdot)$ is strongly measurable; since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map $h$ is $\sigma(Q)$-measurable.

*Set-integral identity.* For $S = Q^{-1}(A)$:

$$\begin{aligned}
\int_S h \, dP &= \int_A \int g \, d\Pi_Q(q, \cdot) \, d(P \circ Q^{-1})(q) \quad \text{(pushforward)} \\
&= \int_{A \times \beta} g(f) \, d\rho(q, f) \quad \text{(disintegration)} \\
&= \int_S g(F) \, dP.
\end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. ☐

**Corollary 2.4** (Predictive sufficiency)**.** $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ *$P$-a.e.*

*Proof.* $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$-measurable by the theorem. ☐

The operator $(K_{Q_*} g)(q_*) = \int g \, dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*} g) \circ Q_*$ $P$-a.e.

## 3 Dynamics and duality

### 3.1 Markov semigroup

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ be a Markov kernel on the state space $\gamma$, with $\Pi_0(q, \cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q, \cdot) \;=\; \int_\gamma \Pi_t(q', \cdot) \, d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s)$$

is a *Markov semigroup kernel.* The associated *Markov operators* are

$$(K_t g)(q) \;:=\; \int g \, d\Pi_t(q, \cdot), \qquad g : \gamma \to \mathbb{R}.$$

Each $K_t$ is positive, unital, and contractive on bounded measurable functions, with $K_0 = \mathrm{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

**Proposition 3.1** (Semigroup property). *$K_{t+s}g = K_t(K_s g)$ for all bounded measurable $g$.*

*Proof.* $(K_{t+s}g)(q) = \int g\, d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \int \left(\int g\, d\Pi_t(q', \cdot)\right) d\Pi_s(q, q') = \int (K_t g)(q')\, d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. □

The pushforward of a measure $\mu$ on $\gamma$ through the kernel is $P_t^*\mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot)\, d\mu(q)$.

## 3.2 Koopman–Perron duality

**Theorem 3.2** (Koopman–Perron duality). *For every bounded measurable $g$ and finite measure $\mu$:*

$$\int (K_t g)\, d\mu \;=\; \int g\, d(P_t^*\mu).$$

*Proof.* $\int g\, d(P_t^*\mu) = \int g\, d(\mu \star \Pi_t) = \int_\gamma \int g\, d\Pi_t(q, \cdot)\, d\mu(q) = \int (K_t g)\, d\mu$, by Bochner–Fubini. □

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \to \gamma$, the stochastic picture collapses.

## 3.3 Specialisation to measure-preserving systems

**Proposition 3.3** (Deterministic limit). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

*Proof.* $(K_t g)(q) = \int g\, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. □

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g\, d\mu = \int g\, d(P_t^*\mu)$ becomes the change-of-variables formula $\int g \circ T^t\, d\mu = \int g\, d(T_*^t\mu)$.

**Example 3.4** (Finite Markov chain). Let $X = \{1, \ldots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution $\mu$ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The predictive kernel is $\Pi(i, \cdot) = P(i, \cdot)$, the $i$-th row of $P$: this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded $f$. The $t$-step kernel is $\Pi^{(t)}(i, \cdot) = P^t(i, \cdot)$, and temporal coherence gives $\Pi^{(t+s)} = \Pi^{(t)} \circ_\kappa \Pi^{(s)}$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g\, d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g\, d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically $\sigma$-additive, so the admissibility condition of [5] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever $h$ separates points of $X$.

The full structure produced by the prediction construction is captured in one definition.

**Definition 3.5** (Observable dynamical system). An *observable dynamical system* is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of $Q_*$), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of $P$ along $Q_*$, and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

## 4 From prediction to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n\omega))_{n\in\mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The predictive state space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that $h$ can see. Whether it is a faithful image of $X$ depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of $X$. Injectivity of $\Phi_h$ modulo $\mu$-null sets is equivalent to the observable $\sigma$-algebra

$$\mathcal{O}_h = \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big)$$

equalling $\mathcal{B}(X)$ modulo $\mu$.

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable $\sigma$-algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$. The next two sections answer this question.

### 4.1 Separation defect and the role of CE

Before specialising to the delay-map setting, the general form of the reconstruction question and the role CE plays in it deserve to be named. Given any finite family of queried levels $n_1, \dots, n_k$ from a directed query system $(\mathcal{E}_n)_n$ with realization measure $\mu$ on $\Omega$, define the *separation defect* of the joined algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$ by

$$\delta(\mathcal{G}) \;:=\; \sup_{A\in\sigma(\mathcal{C})} \inf_{E\in\mathcal{G}} \mu(A \triangle E),$$

where $\sigma(\mathcal{C})$ is the observable $\sigma$-algebra generated by the full query system. This quantity measures how much of the observable structure remains invisible to the finite queried subfamily: $\delta(\mathcal{G}) = 0$ if and only if $\sigma(\mathcal{G}) = \sigma(\mathcal{C})$ modulo $\mu$.

A sequence $(\mathcal{G}_k)$ of joined algebras is *honest* if it is increasing and exhaustive: $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C})$ modulo $\mu$. Under an honest refinement scheme, the separation defect is non-increasing in $k$.

**Proposition 4.1** (CE drives separation defect to zero)**.** *Let $(\mathcal{E}_n)_n$ be a query system with realization measure $\mu$ satisfying collective exhaustion [5]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \;\longrightarrow\; 0 \quad \text{as } k \to \infty.$$

*Proof.* Fix $\varepsilon > 0$ and $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $A$ is approximable modulo $\mu$ by some $E_k \in \mathcal{G}_k$ for all large $k$: the sets $A \triangle E_k$ decrease to a $\mu$-null set. CE [5] rules out persistent mass on globally vanishing events, so $\mu(A \triangle E_k) \to 0$. Taking the supremum over $A$ and the infimum over $E_k$ gives $\delta(\mathcal{G}_k) \to 0$. □

**Remark 4.2** (The temporal specialisation)**.** In the measure-preserving setting with $Q_t = h \circ T^t$, the finite joined algebra $\mathcal{O}_h^{(L)} = \sigma(h, h \circ T, \dots, h \circ T^L)$ forms an increasing family, and under reconstruction ($\mathcal{O}_h = \mathcal{B}$ mod $\mu$, established in §5–6) it is exhaustive — hence an honest refinement scheme. The separation defect $\delta(\mathcal{G})$ specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$, the quantity that drives the finite-sample theory of [4]. There, exponential decay of $\delta(L)$ under mixing yields minimax-optimal certification rates; the honest-refinement condition is automatic because lag growth is the canonical exhaustion mechanism. In the general query setting, these two roles — exhaustion and mixing — must be supplied separately.

# 5 Density bridge and reconstruction

## 5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \to X$ is a measurable, $\mu$-preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

**Definition 5.1** (Observable algebra)**.** The *observable algebra* generated by $h$ is

$$\mathcal{O}_h \; := \; \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big) \; \subseteq \; \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

**Definition 5.2** (Delay map)**.** The *delay map* is

$$\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}, \qquad \Phi_h(x) \; = \; \big(h(T^n x)\big)_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection $\pi_n$, the map $\Phi_h$ is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that $\Phi_h$ is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the $L^2$ level. It holds for any subalgebra of $L^\infty$.

**Lemma 5.3** (Density bridge)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function* 1*. Then:*

$$\overline{\mathcal{A}}^{L^2} \; = \; L^2(X, \, \sigma(\mathcal{A}), \, \mu).$$

*Consequently,* $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ *if and only if* $\sigma(\mathcal{A}) = \mathcal{B}$ *modulo* $\mu$*.*

*Proof.* Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

**Step 1:** $V \subseteq L^2(X, \mathcal{M}, \mu)$**.** Every $f \in \mathcal{A}$ is $\mathcal{A}$-measurable, hence $\mathcal{M}$-measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

**Step 2:** $L^2(X, \mathcal{M}, \mu) \subseteq V$**.** It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in $V$. By the functional monotone class theorem [1], the class

$$\mathcal{H} \; := \; \big\{ f \in L^\infty(X, \mathcal{M}, \mu) : f \in V \big\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \to f$ pointwise a.e. with $f \in L^\infty$, then $f_n \to f$ in $L^2$ by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$-indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$-simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and $V$ is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

**Conclusion.** Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. □

**Remark 5.4** (Role of the algebra structure)**.** The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_{\mathcal{A}} = \overline{\mathrm{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu =$ Lebesgue, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a $\sigma$-algebra (the dyadic structure separates points). But $\mathrm{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain $x^2$. So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, $h$ cyclic for $U_T$ (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 6.2.

### 5.2 Reconstruction theorem

**Theorem 5.5** (Reconstruction theorem)**.** *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

*(i)* $\mathcal{O}_h = \mathcal{B}$ *modulo* $\mu$*;*

*(ii)* $\mathcal{A}_h$ *is dense in* $L^2(X, \mu)$*;*

*(iii)* $\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}$ *is a measure-theoretic embedding:* $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ *modulo* $\mu$*.*

*Proof. (i)* $\Leftrightarrow$ *(ii).* Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B}$ mod $\mu$.

*(i)* $\Leftrightarrow$ *(iii).* By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B}$ mod $\mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ mod $\mu$, which is exactly condition (iii). □

**Remark 5.6** (What "measure-theoretic embedding" means)**.** Condition (iii) says that $\Phi_h$ is *almost everywhere injective* and the pullback $\sigma$-algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ mod $\mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^{\mathbb{Z}}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, $\Phi_h$ induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$.

The $T$-invariance of $\mu$ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift $\sigma$ on $\mathbb{R}^{\mathbb{Z}}$, since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ for each $n$. The dynamical system $(X, T, \mu)$ is therefore isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of $\Phi_h$.

## 6 Cyclic vectors and Stone space identification

### 6.1 Cyclic vectors

**Definition 6.1** (Cyclic vector)**.** The function $h \in L^2(X, \mu)$ is a *cyclic vector* for $U_T$ if

$$\mathcal{H}_h \;:=\; \overline{\operatorname{span}}\{U_T^n h : n \in \mathbb{Z}\} \;=\; L^2(X, \mu).$$

**Corollary 6.2** (Cyclic implies reconstruction)**.** *If $h \in L^\infty(X, \mu)$ is a cyclic vector for $U_T$, then $\mathcal{O}_h = \mathcal{B}$ modulo $\mu$.*

*Proof.* Since $\mathcal{A}_h$ contains $\operatorname{span}\{h \circ T^n\} = \operatorname{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B}$ mod $\mu$. □

**Remark 6.3** (The implication is strict)**.** The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply $h$ is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded $h$, the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when $U_T$ has *simple $L^2$-spectrum*: $L^2(X, \mu)$ is a cyclic $U_T$-module, meaning there exists some cyclic vector. In this case, every generating observation $h$ (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [2] but is not universal.

### 6.2 Stone space identification

The Stone space $\operatorname{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \to \operatorname{St}(\mathcal{O}_h)$ sending $x$ to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

**Proposition 6.4** (Stone space identification)**.** *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo $\mu$. Then $\iota$ is a measure-theoretic isomorphism: $\iota$ identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\mathrm{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\mathrm{St}(\mathcal{O}_h))/\iota_*\mu)$.*

*Proof. Injectivity mod $\mu$.* Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\ni x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing $x'$ in a null set. Hence $\iota$ is injective $\mu$-a.e.

*$\sigma$-algebra identification.* The pullback $\iota^{-1}$ maps the Baire $\sigma$-algebra of $\mathrm{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B}$ mod $\mu$.

*Measure recovery.* Since $\mu$ is $\sigma$-additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [5] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space [5] — built as a compact ambient space for deriving $\sigma$-additivity — is, under the reconstruction condition, the state space $X$ itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

**Example 6.5** (Cantor set: reconstruction without smoothness)**.** Let $\mathcal{C} \subset [0,1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \ldots) = \sum_{k\geq 1} d_k/3^k$. Let $S : \mathcal{C} \to \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0,2\}^{\mathbb{N}}, \nu, \sigma)$ via $\pi$; in particular $S$ is $\mu_{\mathcal{C}}$-invariant and ergodic.

For $x = \pi(d_1, d_2, \ldots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \ldots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in \big[0, \tfrac{1}{3}\big], \qquad d_{n+1} = 2 \iff S^n x \in \big[\tfrac{2}{3}, 1\big],$$

so $h \circ S^n$ recovers the $(n+1)$-st ternary digit of $x$. The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \ldots, S^L x)$ reads off digits $d_1, \ldots, d_{L+1}$; its fibres are the generation-$(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k,\ 1 \leq k \leq L+1\}, \quad z \in \{0,2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0,2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$-a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \to \infty$:

$$\mathcal{O}_h \;=\; \sigma\Big(\bigcup_{L\geq 0} \mathcal{O}_h^{(L)}\Big) \;=\; \mathcal{B}(\mathcal{C}) \quad \mathrm{mod}\ \mu_{\mathcal{C}}.$$

By Theorem 5.5, $\Phi_h$ is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \to \infty$.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2/\log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — $C^2$ ambient dynamics, integer manifold dimension $d$, embedding bound $2d+1$ — are unavailable before one even asks about regularity of $S$. Takens has no domain of application here. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness plays no role.

The classical Takens embedding theorem [7] reaches a related conclusion by a different route: for a $C^2$ diffeomorphism on a compact $d$-manifold and a generic smooth $h$, the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \ldots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d+1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth $h$, the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence $L^2$-dense: the density condition holds and the two results are compatible.

Reconstruction is an exact asymptotic condition: it asks whether the delay algebra generates all of $\mathcal{B}$ in the limit of all lags, and says nothing about rates, finite approximations, or what a finite time series can certify. Those questions are the subject of what follows.

## References


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